Darshan Shirodkar
Pune, MH
Email: darshanshirodkar@gmail.com
Mobile: +91-9665938646

Experience

- **Shopify** — Toronto, ON, Canada
  Senior Data Scientist — *July 2022 - Jun 2026*
  - Partnered with product teams across Shopify's advertising and marketing ecosystem to uncover insights on product adoption, usage, and performance, helping shape product direction.
  - Worked closely with cross-functional stakeholders to design and execute A/B experiments, validate hypotheses, and translate analytical findings into actionable product insights.
  - Owned data modeling and pipeline migration initiatives using PySpark and DBT, delivering analytical capabilities and reporting to Shopify's merchants.
  - Developed attribution and performance measurement models for 1P and 2P ads, helping merchants better understand advertising performance and allocate budgets across ad types.
  - Worked extensively across the GCP data ecosystem, including BigQuery, Composer, and Dataflow, to build scalable data and analytics solutions.

- **Zendrive** — Bengaluru, India
  Senior Data Scientist — *Jan 2021 - Jun 2022*
  - Served as the lead data scientist for the Public Transport Detection team, guiding development of algorithms based on map matching and vehicular dynamics to classify trips as "car drive" or "public transit."
  - Collaborated with engineering and product stakeholders to translate business requirements into machine learning solutions for usage-based and driving-behavior-based insurance applications.
  - Mentored junior data scientists on machine learning methodologies, problem solving, and project execution, providing technical guidance and supporting their professional development.

- **Microsoft** — Bengaluru, India
  Senior Data Scientist — *Dec 2018 - Jan 2021*
  - Partnered with engineering and product teams on optimization of ranking, allocation, and pricing of sponsored search ads across non-US markets for the Bing Search Sponsored Ads team.
  - Developed a real-time reinforcement learning algorithm for adaptive ad layout optimization, enabling the search results page layout to dynamically respond to changing conditions.
  - Shaped the feature engineering strategy for a deep neural network used to estimate ad relevance and rank ads in real-time auctions.

- **Precision Wellness** — San Francisco, CA
  Data Scientist - Consultant — *Feb 2018 - Nov 2018*
  - Developed a statistical model for predicting cardiovascular disease risk from patient risk factors and scaled the implementation to millions of patient records using PySpark.

- **Adobe** — Bengaluru, India
  Data Science Manager — *Sep 2015 - Oct 2017*
  - Managed a team of data scientists focused on Adobe Media Optimizer, a search and display advertising bidding platform, providing technical direction and supporting execution of data science initiatives.
  - Developed a "bump hunting" algorithm to categorize bid requests into a tractable number of significant groups, creating a foundation for more targeted bid optimization.
  - Guided development of bid-to-cost, bid-to-impressions, and bid-to-revenue models for each group, along with a convex optimization formulation for the bid optimizer.

- Coordinated the team's work on real-time estimation and adaptive control algorithms for the display ad bidder, balancing modeling accuracy with real-time system requirements.
- Built a Python simulation environment for evaluating real-time bidding strategies and enabling experimentation with online auction strategies.

- **Pubmatic** Pune, India
  Data Scientist *June 2013 - Sep 2015*
  - Served as a technical lead within the horizontal machine learning team, tackling revenue and infrastructure optimization problems and collaborating across teams on high-impact initiatives.
  - Developed a classification model to filter low-quality ad opportunities from billions of ad requests, reducing infrastructure usage by 12%.

- **TCS Innovation Lab** Pune, India
  Research Engineer *Nov 2011 - Jun 2013*
  - Applied statistical modeling and machine learning techniques to diverse problems across quantitative finance and human resources.
  - Developed R-based simulations demonstrating the efficiency of Multi-level Monte Carlo techniques for financial option pricing.
  - Collaborated with the HR department to analyze employee data and identify groups with high attrition risk.

Education

- **University of Illinois at Urbana-Champaign** Urbana, IL
  Master of Science in Applied Mathematics; GPA: 3.96 *2009 – 2011*
- **Indian Institute of Science** Bengaluru, India
  Master of Engineering in Signal Processing; GPA: 7.4/8 *2006 – 2008*
- **KJ somaiya College of Engineering** Mumbai, India
  Bachelor of Engineering in Electronics *2002 – 2006*

Publications

- D. Shirodkar and S. Meyn,"Quasi stochastic approximation," Proceedings of the 2011 American Control Conference, San Francisco, CA, USA, 2011, pp. 2429-2435, doi: 10.1109/ACC.2011.5991485.

Skills

- **Languages**: Python, SQL
- **Tools & Technologies**: Pyspark, Hadoop, Google Cloud Services, Bigquery, Looker Studio